Risk-Aware Adaptive Extended Kalman Filtering for Autonomous Cislunar Spacecraft Navigation

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April 29, 2026

Abstract

Autonomous spacecraft navigation in cislunar space is an active aerospace problem because spacecraft cannot always rely on continuous ground support or Earth-orbit navigation infrastructure. This paper proposes a laptop-scale computational study of a risk-aware adaptive Extended Kalman Filter (EKF) for planar Earth-Moon cislunar navigation. The spacecraft dynamics are modeled with the circular restricted three-body problem (CR3BP), and range-plus-bearing measurements are simulated under clean and stressed conditions. The proposed navigation risk score combines covariance growth, measurement geometry, missed-measurement events, and lighting/visibility penalties. The score adapts process or measurement noise and is compared with a fixed-noise baseline EKF.

1 Introduction

Cislunar navigation is becoming increasingly important as lunar and Earth-Moon-space missions expand. Current research explores autonomous onboard navigation, optical navigation, cooperative ranging, and robust filtering for spacecraft operating beyond ordinary GPS coverage [3, 1, 2].

This project asks a focused computational question: can an interpretable navigation risk score improve EKF robustness during missed measurements, poor measurement geometry, and lighting or visibility degradation?

2 CR3BP Dynamics

The planar CR3BP state is

$$\boldsymbol{x} = [x, y, \dot{x}, \dot{y}]^T. \tag{1}$$

Using the Earth-Moon mass parameter μ, the nondimensional rotating-frame equations are

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3}, \tag{2}$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}, \tag{3}$$

where

$$r_1 = \sqrt{(x+\mu)^2 + y^2}, \qquad r_2 = \sqrt{(x-1+\mu)^2 + y^2}. \tag{4}$$

The simulator propagates truth and EKF predictions with fourth-order Runge-Kutta integration.

3 Measurement Model

The first measurement model uses range and bearing from a reference observer:

$$\boldsymbol{z} = \begin{bmatrix} \rho \\ \theta \end{bmatrix} = \begin{bmatrix} \sqrt{(x-x_o)^2 + (y-y_o)^2} \\ \tan^{-1}\left((y-y_o)/(x-x_o)\right) \end{bmatrix} + \boldsymbol{\eta}. \tag{5}$$

The measurement Jacobian is analytic and the measurement noise covariance is $\boldsymbol{R}$.

4 Baseline EKF

The EKF prediction is

$$\hat{\boldsymbol{x}}_{k|k-1} = f(\hat{\boldsymbol{x}}_{k-1|k-1}), \tag{6}$$

$$\boldsymbol{P}_{k|k-1} = \boldsymbol{\Phi}_k \boldsymbol{P}_{k-1|k-1} \boldsymbol{\Phi}_k^T + \boldsymbol{Q}. \tag{7}$$

The measurement update is

$$\boldsymbol{K}_k = \boldsymbol{P}_{k|k-1} \boldsymbol{H}_k^T (\boldsymbol{H}_k \boldsymbol{P}_{k|k-1} \boldsymbol{H}_k^T + \boldsymbol{R})^{-1}, \tag{8}$$

$$\hat{\boldsymbol{x}}_{k|k} = \hat{\boldsymbol{x}}_{k|k-1} + \boldsymbol{K}_k(\boldsymbol{z}_k - \hat{\boldsymbol{z}}_k), \tag{9}$$

$$\boldsymbol{P}_{k|k} = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k) \boldsymbol{P}_{k|k-1} (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k)^T + \boldsymbol{K}_k \boldsymbol{R} \boldsymbol{K}_k^T. \tag{10}$$

5 Navigation Risk Score

The proposed risk score is

$$\boxed{\mathcal{R}_{\text{nav}}(t) = w_r \frac{\sigma_r(t)}{r_0} + w_v \frac{\sigma_v(t)}{v_0} + w_g \frac{1}{\Gamma(t) + \epsilon} + w_m M(t) + w_l L(t)} \tag{11}$$

where $\sigma_r = \sqrt{\operatorname{tr}(\boldsymbol{P}_{rr})}$, $\sigma_v = \sqrt{\operatorname{tr}(\boldsymbol{P}_{vv})}$, Γ is a measurement geometry strength score, M is a missed-measurement flag, and L is a lighting/visibility penalty.

6 Adaptive EKF Rules

The risk score is used in two adaptive filters:

$$\boldsymbol{R}(t) = \boldsymbol{R}_0 \left[1 + \beta \operatorname{clip}(\mathcal{R}_{\text{nav}}, 0, R_{\max})\right], \tag{12}$$

$$\boldsymbol{Q}(t) = \boldsymbol{Q}_0 \left[1 + \alpha \operatorname{clip}(\mathcal{R}_{\text{nav}}, 0, R_{\max})\right]. \tag{13}$$

The clipping prevents unbounded noise inflation.

7 Experiments

The planned experiments compare a baseline EKF, a risk-adaptive measurement EKF, and a risk-adaptive process EKF across clean and stressed cases: missed measurements, bad geometry, lighting loss, and combined stress. The main diagnostics are RMS position error, maximum position error, final velocity error, covariance growth, maximum risk score, and missed-measurement fraction.

8 Preliminary Results

The first controlled scenario suite separated clean navigation, missed measurements, bad geometry, lighting degradation, and a combined-stress case. Table 1 summarizes the best filter in each case. The results show that risk adaptation is not universally beneficial. It is most useful when the estimator is actually stressed by measurement loss or geometry, while the fixed baseline remains competitive in clean and lighting-only cases.

Table 1: First-pass controlled scenario results. RMS errors are nondimensional CR3BP position errors from deterministic single-seed runs.

Scenario	Best filter	Baseline RMS	Best RMS	Improvement
Clean baseline	Baseline	4.232×10^{-4}	4.232×10^{-4}	0.0%
Missed measurements	Adaptive $\boldsymbol{R}$	2.470×10^{-1}	3.632×10^{-3}	98.5%
Bad geometry	Adaptive $\boldsymbol{R}$	4.221×10^{-4}	3.075×10^{-4}	27.2%
Lighting loss	Baseline	5.636×10^{-4}	5.636×10^{-4}	0.0%
Combined stress	Adaptive $\boldsymbol{Q}$	1.005×10^{-3}	8.504×10^{-4}	15.4%

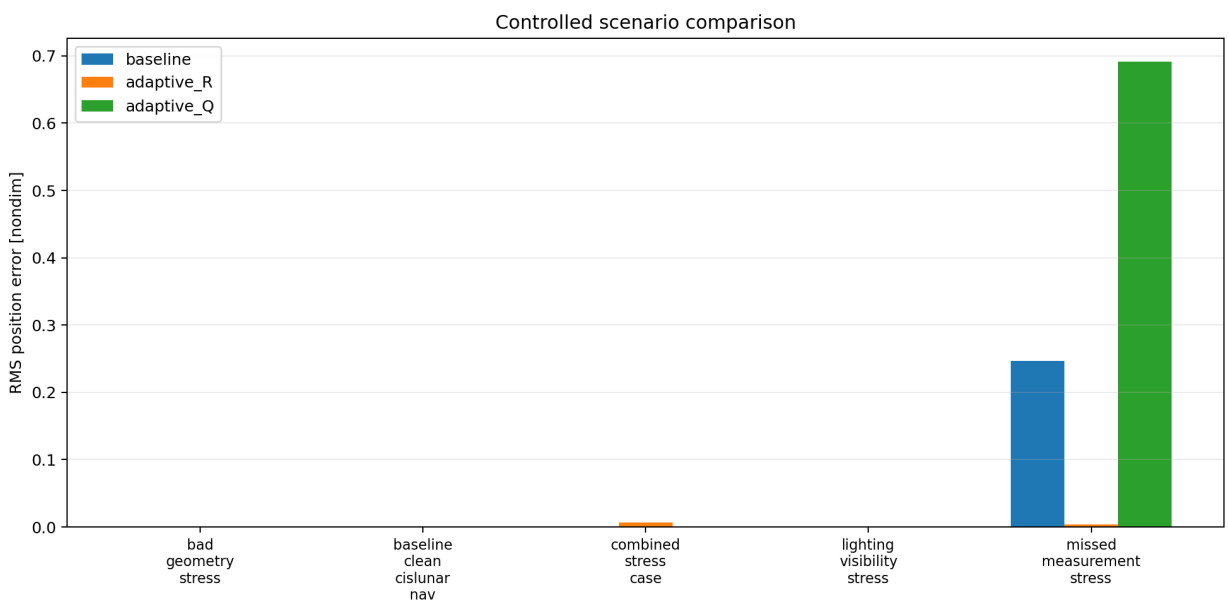

Figure 1: RMS position-error comparison across controlled scenarios. The adaptive filters help in some stress cases, but the baseline remains best or tied in clean and lighting-only cases.

The first implemented parameter scan used a combined-stress scenario with missed measurements, lighting degradation, and weak geometry. The scan varied α, β, and w_g over 81 filter summaries. These results are preliminary; their purpose is to test whether the simulation and comparison pipeline can expose useful estimator behavior.

The fixed-noise baseline EKF produced an RMS position error of approximately 1.24 nondimensional distance units in the combined-stress scan. The best adaptive measurement-noise case used $\beta = 0.6$ and $w_g = 0.5$, reducing RMS position error to approximately 0.589, a 52.5% improvement over the same-run baseline. However, more aggressive settings were not uniformly beneficial. For

example, $\beta = 1.0$ with $w_g = 0.3$ increased RMS position error to approximately 1.48 and raised maximum position error to approximately 9.42. This confirms that risk-aware adaptation requires parameter discipline; it is not automatically superior to a standard EKF.

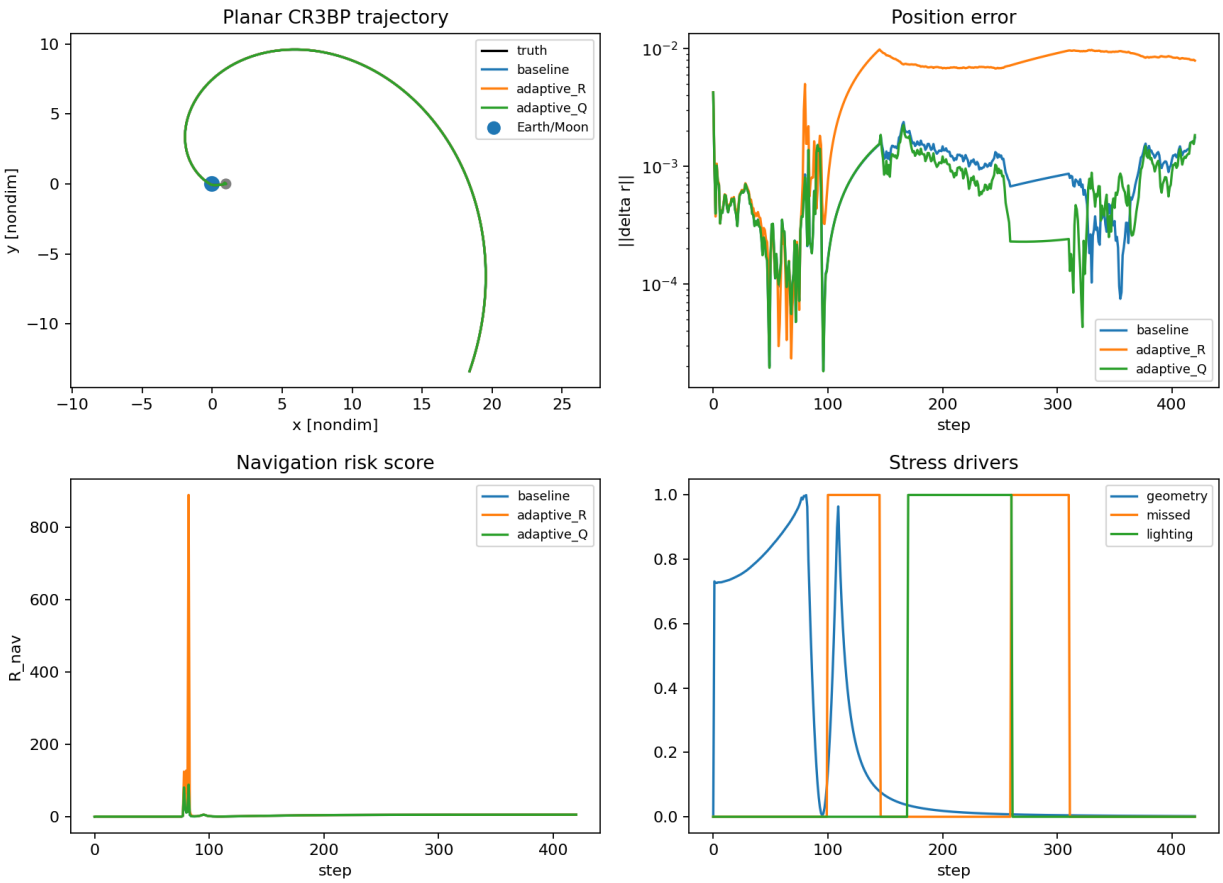

Figure 2: Combined-stress run showing trajectory, position error, navigation risk, and stress-driver histories.

The raw risk score can become very large when the geometry-strength term Γ approaches a weak-information regime. This is expected from the reciprocal form $1/(\Gamma + \epsilon)$, but it also means that raw risk values should be interpreted as diagnostic signals, not direct noise multipliers. The implemented filters use $\mathrm{clip}(\mathcal{R}_{\mathrm{nav}}, 0, R_{\max})$ before adapting $\boldsymbol{Q}$ or $\boldsymbol{R}$. The next research step is therefore not simply to maximize risk response, but to study clipped risk, geometry normalization, and failure-mode-specific tuning.

9 Limitations

This is a v1 computational model. It uses planar CR3BP rather than full ephemeris dynamics, simplified observers, and simplified visibility penalties. The purpose is to test estimator behavior under controlled assumptions, not to claim flight-ready navigation performance. The current result also shows that the risk equation can be over-sensitive to weak geometry unless its scale and clipping are treated carefully. This sensitivity is not a reason to discard the approach; it is a core research target for the next experiments.

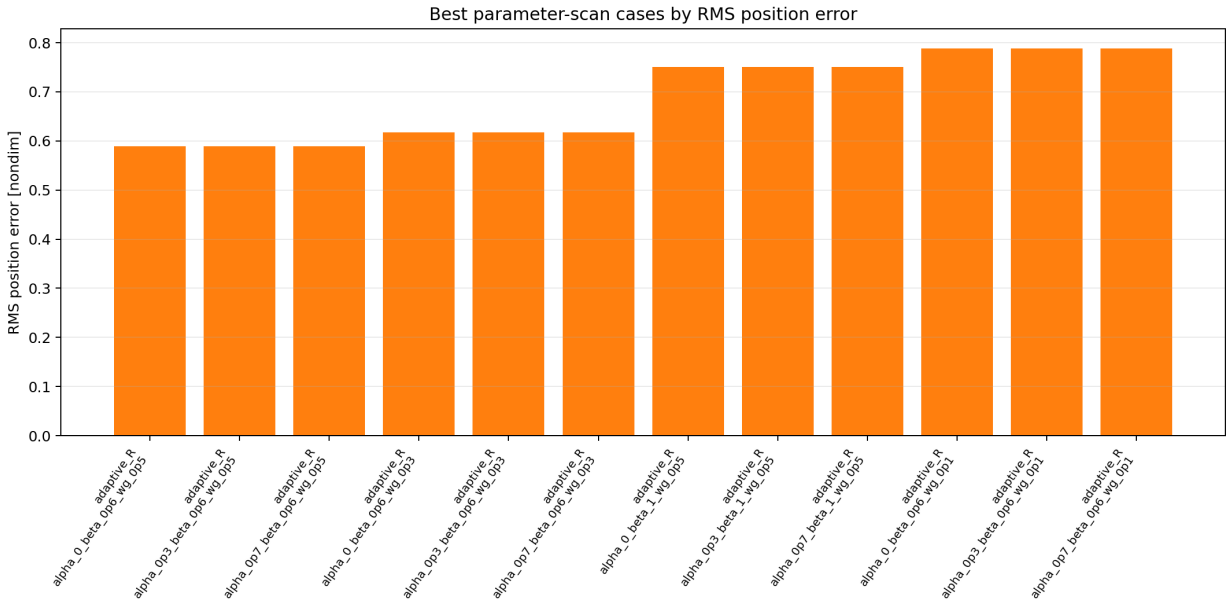

Figure 3: Best parameter-scan cases ranked by RMS position error. The best early cases use adaptive measurement-noise inflation rather than the fixed baseline.

10 Conclusion

Risk-aware adaptive filtering is a defensible aerospace research direction because it connects estimator uncertainty to operational stressors. The value of this project depends on reproducible tests showing when the risk score helps, when it fails, and whether it improves robustness over a fixed-noise EKF.

References

[1] NASA. Nasa optical navigation tech could streamline planetary exploration, 2024. URL `https://www.nasa.gov/technology/goddard-tech/nasa-optical-navigation-tech-could-streamline-planetary-exploration/`.

[2] Oguri Research Group. Autonomous navigation in cislunar space, 2025. URL `https://engineering.purdue.edu/OguriGroup/research/auto-nav`.

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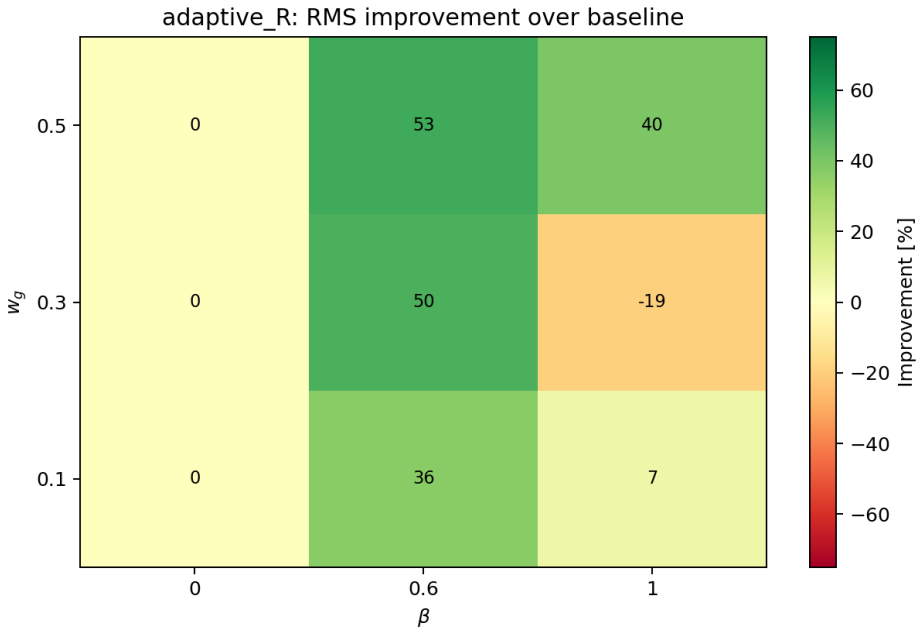

Figure 4: Adaptive measurement-noise RMS improvement relative to the baseline EKF. Green cells indicate improvement; red cells indicate degradation.